**PRODUCT REQUIREMENT DOCUMENTATION (PRD)**

# UMHackathon 2026

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## TEAM NAME: Its Works On My Machine

## TEAM CODE: 106


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# 1. Project Overview

## 1.1 Problem Statement

In modern organizations, especially those with multiple branch offices, daily operations depend on communication between Headquarters (HQ) and Branch Offices. However, this communication is often fragmented across different tools such as emails, spreadsheets, messaging platforms, and reports.

As a result:

- Information is scattered and difficult to understand
- Many workflows require manual interpretation and decision-making
- Processes are slow, inconsistent, and prone to human error
- Decision-making is delayed due to back-and-forth communication
- Unstructured data (e.g. news, feedback, reports) cannot be effectively used

Traditional automation systems are limited because they:

- Rely on fixed rules
- Cannot think step-by-step
- Cannot handle unclear or missing data
- Cannot manage complex workflows dynamically

This creates a major gap where organizations cannot respond quickly, coordinate efficiently, or make data-driven decisions in real time.



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## 1.2 Target Domain

The SynapseCore System is developed under the domain of AI Systems & Agentic Workflow Automation.

The system focuses on enabling:

- Intelligent workflow orchestration
- AI-driven multi-step reasoning
- Automated decision-making processes
- Real-time coordination between HQ and Branch Offices

It is applicable in enterprise environments that require:

- Continuous data exchange
- Complex workflow coordination
- Real-time responsiveness
- Integration with APIs and data systems

## 1.3 Proposed Solution Summary

SynapseCore System is a web-based AI-powered workflow platform that transforms unstructured operational inputs into structured, actionable workflows.

The system operates using a closed-loop workflow model:
Input → Project → Phase → Plan → Validate → Execute → Improve

Key capabilities include:

- Understanding unstructured inputs such as messages, reports, and documents
- Converting inputs into structured projects
- Generating multi-step workflow plans
- Validating information and reducing errors (hallucination limiter)
- Executing workflows through system actions
- Continuously improving workflows based on feedback and outcomes

At the core of the system, Z.AI GLM (General Language Model) acts as the central reasoning engine, enabling intelligent decision-making and adaptive workflow execution.

# 2. Background & Business Objective

## 2.1 Background of the Problem

In modern enterprise environments, operations are distributed across multiple branches and departments. Coordination between HQ and Branch Offices is essential but often inefficient due to disconnected systems and manual processes.

Most workflows involve:

- Collecting data from multiple sources
- Interpreting information manually
- Making decisions based on incomplete data
- Executing tasks through separate tools

Additionally, much of the data is unstructured, including:

- Market trends
- Customer feedback
- External news
- Operational reports

Traditional systems cannot effectively process or utilize this type of data, leading to delays, inefficiencies, and missed opportunities.

## 2.2 Importance of Solving This Issue

Solving this problem is critical to:

- Improve operational efficiency
- Enable faster and more accurate decision-making
- Reduce dependency on manual processes
- Enhance coordination between HQ and branches

By automating workflows and integrating AI-driven reasoning, organizations can:

- Respond quickly to real-time changes
- Optimize inventory and sales strategies
- Improve overall system reliability



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## 2.3 Strategic Fit and Impact

The SynapseCore System aligns with industry trends toward:

- AI-powered automation
- Intelligent workflow systems
- Data-driven decision-making

Its impact includes:

- Transforming fragmented workflows into structured processes
- Reducing communication gaps
- Enabling scalable and adaptive enterprise operations


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# 3. Product Purpose

## 3.1 Main Goal of the System

The main goal of the SynapseCore System is to provide an intelligent workflow orchestration platform that connects data, decision-making, and execution into a single automated system.

Specifically, the system aims to:

- Reduce communication gaps between HQ and Branch Offices
- Enable smooth and efficient task handovers
- Automate multi-step workflows using AI
- Improve decision speed and quality

## 3.2 Intended Users

- Headquarters (HQ) for decision-making and monitoring
- Branch Offices for submitting requests and executing workflows
- Operations managers and business stakeholders



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# 4. System Functionalities

## 4.1 System Description

The SynapseCore System operates as an AI-powered agentic workflow engine that continuously processes inputs, generates decisions, and executes actions.

System workflow:

- Receive unstructured input from users or systems
- Use AI (GLM) to understand and extract key information
- Convert input into a structured project
- Break project into phases
- Generate workflow plans
- Validate plans and information
- Execute actions through integrated systems
- Monitor outcomes and improve future workflows

The system is:

- Stateful (tracks workflow progress)
- Adaptive (responds to new inputs and changes)
- Intelligent (uses AI for reasoning and decision-making)

## 4.2 Core Functionalities

- AI-powered understanding of unstructured inputs
- Project creation and workflow monitoring
- Multi-phase workflow management
- AI-generated planning and decision-making
- Validation system to reduce hallucination and errors
- Workflow execution through APIs and system actions
- Human-in-the-loop approval mechanism
- Real-time monitoring and reporting

## 4.3 AI Model and Prompt Design

### 4.3.1 Model Selection

The system uses Z.AI GLM (General Language Model) as the central reasoning engine. The model is responsible for:

- Understanding input data
- Performing multi-step reasoning
- Generating structured outputs
- Guiding workflow decisions

Without the AI model, the system would not be able to perform intelligent workflow coordination.

### 4.3.2 Prompting Strategy

The system uses a multi-step agentic prompting strategy, where AI performs reasoning across different workflow stages.

Steps include:

- Understanding input
- Identifying key problems
- Generating project structure
- Planning workflow phases
- Validating decisions
- Producing structured outputs

Prompts are designed to ensure:

- Logical reasoning
- Reduced hallucination
- Consistent and reliable outputs

#### 4.3.3 Context and Input Handling

The system supports multiple types of input:
- Text input (messages, reports)
- Unstructured data (website links)
- Documents (PDF, forms)
- Structured data (inventory, sales records)

The system:
- Extracts relevant information
- Maintains context across workflow phases
- Handles incomplete or ambiguous data

#### 4.3.4 Fallback and Failure Behavior

To ensure reliability, the system includes:
- Retry mechanisms for failed operations
- Validation checks for input data
- Fallback logic for missing information
- Human intervention when necessary



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# 5. User Stories and Use Cases

## 5.1 User Stories

- As a branch office user, I want to submit operational requests easily so HQ can understand and respond quickly
- As an HQ user, I want structured insights to make informed decisions
- As a user, I want AI to guide workflow steps clearly
- As a manager, I want real-time updates and reports

## 5.2 Use Cases

- Inventory planning and allocation
- Sales coordination
- Approval processes
- Reporting and monitoring
- Risk and trend analysis

# 6. Features Included (Scope Definition)

- AI-powered workflow orchestration
- Processing of unstructured inputs
- Project-based workflow management
- Multi-phase execution tracking
- AI planning and validation
- Reporting and analytics
- Approval system

# 7. Features Not Included (Scope Control)

- Full-featured enterprise user interface
- Advanced authentication mechanisms
- Multi-user collaboration features
- Large-scale production deployment infrastructure

# 8. Assumptions and Constraints

## 8.1 LLM Cost Constraint

The use of large language models introduces cost considerations related to token usage. To manage these costs effectively, the system employs optimization strategies such as minimizing prompt size, caching repeated queries, and selectively processing data.

## 8.2 Technical Constraints

The system relies heavily on external APIs for data and functionality. As such, its performance is dependent on the availability and reliability of these services. Network connectivity and API rate limits also play a significant role in determining system performance.

## 8.3 Performance Constraints

System performance is influenced by factors such as API response times and the computational requirements of the AI model. While efforts are made to optimize efficiency, some latency is inherent in real-time processing systems.

## 8.4 User Input Constraints

The accuracy of the system's outputs is contingent upon the quality of the inputs it receives. Users are expected to provide clear and valid data to ensure optimal performance. Ambiguous or incorrect inputs may lead to suboptimal or inaccurate results.


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# 9. Risks and Questions Throughout Development

- Reliability of AI-generated decisions
- Dependency on external APIs and potential disruptions
- Scalability under increasing workloads
- Security concerns related to data handling and API access
- Transparency and explainability of decision-making processes